0.1 Error messages

`\begin{split} won't work here.`
Example:

```
! Package amsmath Error: \begin{split} won't work here.
 ...

l.8 \begin{split}

? h
\Did you forget a preceding \begin{equation}?
If not, perhaps the 'aligned' environment is what you want.
?
```

Explanation: The `split` environment does not construct a stand-alone displayed equation; it needs to be used within some other environment such as `equation` or `gather`.

`Erroneous nesting of equation structures`
Example:

```
! Package amsmath Error: Erroneous nesting of equation structures;
(amsmath)                trying to recover with 'aligned'.

See the amsmath package documentation for explanation.
Type  H <return>  for immediate help.
 ...

l.260 \end{alignat*}
                     \end{equation*}
```

Explanation: The structures `align`, `alignat`, etc., are designed for top-level use and for the most part cannot be nested inside some other displayed equation structure. The chief exception is that `align` and most of its variants can be used inside the `gather` environment.

`Extra & on this line`
Example:

```
! Package amsmath Error: Extra & on this line.

See the amsmath package documentation for explanation.
Type  H <return>  for immediate help.
 ...

l.9 \end{alignat}

? h
\An extra & here is so disastrous that you should probably exit
 and fix things up.
?
```

Explanation: In an `alignat` structure the number of alignment points per line is dictated by the numeric argument given after `\begin{alignat}`. If you use more alignment points in a line it is assumed that you accidentally left out a newline command `\\` and the above error is issued.

Font OMX/cmex/m/n/7=cmex7 not loadable ...

Example:

```
! Font OMX/cmex/m/n/7=cmex7 not loadable: Metric (TFM) file not found.
<to be read again>
                   relax
l.8 $a
      b+b^2$
? h
I wasn't able to read the size data for this font,
so I will ignore the font specification.
[Wizards can fix TFM files using TFtoPL/PLtoTF.]
You might try inserting a different font spec;
e.g., type `I\font<same font id>=<substitute font name>'.
?
```

Explanation: Certain extra sizes of some Computer Modern fonts that were formerly available mainly through the AMSFonts distribution are considered part of standard LaTeX (as of June 1994): `cmex7`–`9`, `cmmib5`–`9`, and `cmbsy5`–`9`. If these extra sizes are missing on your system, you should try first to get them from the source where you obtained LaTeX. If that fails, you could try getting the fonts from CTAN (e.g., in the form of Metafont source files, directory `/tex-archive/fonts/latex/mf`, or in PostScript Type 1 format, directory `/tex-archive/fonts/cm/ps-type1/bakoma`).

If the font name begins with `cmex`, there is a special option `cmex10` for the `amsmath` package that provides a temporary workaround. I.e., change the `\usepackage` to

```
\usepackage[cmex10]{amsmath}
```

This will force the use of the 10-point size of the `cmex` font in all cases. Depending on the contents of your document this may be adequate.

Improper argument for math accent

Example:

```
! Package amsmath Error: Improper argument for math accent:
(amsmath)                Extra braces must be added to
(amsmath)                prevent wrong output.

See the amsmath package documentation for explanation.
Type  H <return>  for immediate help.
 ...

l.415 \tilde k_{\lambda_j} = P_{\tilde \mathcal
                                                {M}}
?
```

Explanation: Non-simple arguments for any LaTeX command should be enclosed in braces. In this example extra braces are needed as follows:

```
... P_{\tilde{\mathcal{M}}}
```

`Math formula deleted: Insufficient extension fonts`
Example:

```
! Math formula deleted: Insufficient extension fonts.
l.8 $ab+b^2$

?
```

Explanation: This usually follows a previous error `Font ... not loadable`; see the discussion of that error (above) for solutions.

`Missing number, treated as zero`
Example:

```
! Missing number, treated as zero.
<to be read again>
                   a
l.100 \end{alignat}

? h
A number should have been here; I inserted `0'.
(If you can't figure out why I needed to see a number,
look up `weird error' in the index to The TeXbook.)

?
```

Explanation: There are many possibilities that can lead to this error. However, one possibility that is relevant for the amsmath package is that you forgot to give the number argument of an `alignat` environment, as in:

```
\begin{alignat}
 a&  =b&    c& =d\\
a'& =b'&   c'& =d'
\end{alignat}
```

where the first line should read instead

```
\begin{alignat}{2}
```

Another possibility is that you have a left bracket character `[` following a line break command `\\` in a multiline construction such as `array`, `tabular`, or `eqnarray`. This will be interpreted by LaTeX as the beginning of an 'additional vertical space request [**?**, §C.1.6], even if it occurs on the following line and is intended to be part of the contents. For example,

```
\begin{array}{c}
a+b\\
[f,g]\\
m+n
\end{array}
```

To prevent the error message in such a case, you can add braces as discussed in the LaTeX manual [**?**, §C.1.1]:

```
\begin{array}{c}
a+b\\
{[f,g]}\\
m+n
\end{array}
```

or precede the bracketed expression by `\relax`.

`Missing \right. inserted`

Example:

```
! Missing \right. inserted.
<inserted text>
                \right .
l.10 \end{multline}

? h
I've inserted something that you may have forgotten.
(See the <inserted text> above.)
With luck, this will get me unwedged. But if you
really didn't forget anything, try typing '2' now; then
my insertion and my current dilemma will both disappear.
```

Explanation: This error typically arises when you try to insert a line break or `&` between a `\left`-`\right` pair of delimiters in any multiline environment, including `split`.

```
\begin{multline}
AAA\left(BBB\\
  CCC\right)
\end{multline}
```

There are two possible solutions: (1) instead of using `\left` and `\right`, use 'big' delimiters of fixed size (`\bigl \bigr \biggl \biggr` ...; see §**??**); or (2) use null delimiters to break up the `\left`-`\right` pair into parts for each line (or cell):

```
AAA\left(BBB\right.\\
  \left.CCC\right)
```

The latter solution may result in mismatched delimiter sizes; ensuring that they match requires using `\vphantom` in the segment that has the smaller delimiter (or possibly `\smash` in the segment that has the larger delimiter). In the argument of `\vphantom` put a copy of the tallest element that occurs in the other segment, e.g.,

```
xxx \left(\int_t yyy\right.\\
  \left.\vphantom{\int_t} zzz ... \right)
```

`Missing } inserted`
Example:

```
! Missing } inserted.
<inserted text>
                \right .
l.10 \end{multline}

? h
I've inserted something that you may have forgotten.
(See the <inserted text> above.)
With luck, this will get me unwedged. But if you
really didn't forget anything, try typing '2' now; then
my insertion and my current dilemma will both disappear.
```

Explanation: This error is often the result of using `$` within a multiline display environment. Remove such `$` signs (except when they appear within `\text{...}`).

`Old form '\pmatrix' should be \begin{pmatrix}.`
Example:

```
! Package amsmath Error: Old form '\pmatrix' should be
                         \begin{pmatrix}.

See the amsmath package documentation for explanation.
Type  H <return>  for immediate help.
 ...

\pmatrix ->\left (\matrix@check \pmatrix
                                        \env@matrix
l.16 \pmatrix
             {a&b\cr c&d\cr}
? h
'\pmatrix{...}' is old Plain-TeX syntax whose use is
ill-advised in LaTeX.
?
```

Explanation: When the amsmath package is used, the old forms of `\pmatrix`, `\matrix`, and `\cases` cannot be used any longer because of naming conflicts. Their syntax did not conform with standard LaTeX syntax in any case.

`Paragraph ended before \xxx was complete`
Example:

```
Runaway argument?

! Paragraph ended before \multline was complete.
<to be read again>
                  \par
l.100
```

```
? h
I suspect you've forgotten a `}', causing me to apply this
control sequence to too much text. How can we recover?
My plan is to forget the whole thing and hope for the best.
?
```

Explanation: This might be produced by a blank line between the `\begin` and `\end`. Another possibility is a misspelling in the `\end{multline}` command, e.g.,

```
\begin{multline}
...
\end{multiline}
```

or by using abbreviations for certain environments, such as `\bal` and `\eal` for `\begin{align}` and `\end{align}`:

```
\bal
...
\eal
```

For technical reasons that kind of abbreviation does not work with the more complex displayed equation environments of the amsmath package (`gather`, `align`, `split`, etc.; cf. `technote.tex`).

`Runaway argument?`

See the discussion for the error message `Paragraph ended before \xxx was complete`.

`Unknown option `xxx' for package `yyy'`

Example:

```
! LaTeX Error: Unknown option `intlim' for package `amsmath'.
...
? h
The option `intlim' was not declared in package `amsmath', perhaps you
misspelled its name. Try typing  <return>  to proceed.
?
```

Explanation: This means that you misspelled the option name, or the package simply does not have an option that you expected it to have. Consult the documentation for the given package.

0.2 Warning messages

`Cannot use `split' here`

Example:

```
Package amsmath Warning: Cannot use `split' here;
(amsmath)                trying to recover with `aligned'
```

Explanation: The `split` environment is designed to serve as the entire body of an equation, or an entire line of an `align` or `gather` environment. There cannot be any printed material before or after it within the same enclosing structure:

```
\begin{equation}
\left\{ % <-- Not allowed
\begin{split}
...
\end{split}
\right. % <-- Not allowed
\end{equation}
```

- `Foreign command \over [or \atop or \above]`

Example:

```
Package amsmath Warning: Foreign command \over; \frac or \genfrac
(amsmath)                should be used instead.
```

Explanation: The primitive generalized fraction commands of TEX—`\over`, `\atop`, `\above`—are deprecated when the amsmath package is used because their syntax is foreign to LATEX and amsmath provides native LATEX equivalents. See `technote.tex` for further information.

0.3 Wrong output

- Section numbers 0.1, 5.1, 8.1 instead of 1, 2, 3

This most likely means that you have the arguments for `\numberwithin` in reverse order:

```
\numberwithin{section}{equation}
```

That means 'print the section number as *equation number.section number* and reset to 1 every time an equation occurs' when what you probably wanted was the inverse

```
\numberwithin{equation}{section}
```

- The `\numberwithin` command had no effect on equation numbers

Are you looking at the first section in your document? Check the section numbers elsewhere to see if the problem is the one described in the previous item.

- Bracketed material disappears at the beginning of `aligned` or `gathered`

For most multiline equation environments, amsmath disables the LATEX convention to recognize a [bracketed] expression as an option when it occurs at the beginning of an environment or immediately following `\\`, provided a space intervenes. This fails at the beginning of `aligned` and `gathered`, and is a bug. Insert `\relax` before the opening brace to restore the desired result.